

An Analysis of Linear discriminant analysis

Linear discriminant analysis

Linear discriminant analysis -- Part 1

Fisher's linear discriminant The terms Fisher's linear discriminant and LDA are often used interchangeably, although Fisher's original article actually describes a slightly different discriminant, which does not make some of the assumptions of LDA such as normally distributed classes or equal class covariances. Suppose two classes of observations have means $\mu \rightarrow 0, \mu \rightarrow 1$ and covariances Σ_0, Σ_1 . Then the linear combination of features $w \rightarrow T x \rightarrow$ will have means $w \rightarrow T \mu \rightarrow i$ and variances $w \rightarrow T \Sigma_i w \rightarrow$ for $i = 0, 1$. Fisher defined the separation between these two distributions to be the ratio of the variance between the classes to the variance within the classes:

Linear discriminant analysis -- Part 2

Structure Correlation Coefficients: The correlation between each predictor and the discriminant score of each function. This is a zero-order correlation (i.e., not corrected for the other predictors). Standardized Coefficients: Each predictor's weight in the linear combination that is the discriminant function. Like in a regression equation, these coefficients are partial (i.e., corrected for the other predictors). Indicates the unique contribution of each predictor in predicting group assignment. Functions at Group Centroids: Mean discriminant scores for each grouping variable are given for each function. The farther apart the means are, the less error there will be in classification.

Linear discriminant analysis -- Part 3

LDA for two classes Consider a set of observations $x \rightarrow$ (also called features, attributes, variables or measurements) for each sample of an object or event with known class y . This set of samples is called the training set in a supervised learning context. The classification problem is then to find a good predictor for the class y of any sample of the same distribution (not necessarily from the training set) given only an observation $x \rightarrow$. LDA approaches the problem by assuming that the conditional probability density functions $p(x \rightarrow | y = 0)$ and $p(x \rightarrow | y = 1)$ are both the normal distribution with mean and covariance parameters $(\mu \rightarrow 0, \Sigma_0)$ and $(\mu \rightarrow 1, \Sigma_1)$.

, respectively. Under this assumption, the Bayes-optimal solution is to predict points as being from the second class if the log of the likelihood ratios is bigger than some threshold T , so that:

Linear discriminant analysis -- Part 4

where μ is the mean of the class means. The class separation in a direction $w \rightarrow$ in this case will be given by

Linear discriminant analysis -- Part 5

$$\frac{1}{2} (x \rightarrow - \mu \rightarrow 0)^T \Sigma_0^{-1} (x \rightarrow - \mu \rightarrow 0) + \frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} (x \rightarrow - \mu \rightarrow 1)^T \Sigma_1^{-1} (x \rightarrow - \mu \rightarrow 1) - \frac{1}{2} \ln |\Sigma_1| > T$$

Linear discriminant analysis -- Part 6

Practical use In practice, the class means and covariances are not known. They can, however, be estimated from the training set. Either the maximum likelihood estimate or the maximum a posteriori estimate may be used in place of the exact value in the above equations. Although the estimates of the covariance may be considered optimal in some sense, this does not mean that the resulting discriminant obtained by substituting these values is optimal in any sense, even if the assumption of normally distributed classes is correct. Another complication in applying LDA and Fisher's discriminant to real data occurs when the number of measurements of each sample (i.e., the dimensionality of each data vector) exceeds the number of samples in each class. In this case, the covariance estimates do not have full rank, and so cannot be inverted. There are a number of ways to deal with this. One is to use a pseudo inverse instead of the usual matrix inverse in the above formulae. However, better numeric stability may be achieved by first projecting the problem onto the subspace spanned by Σ_b . Another strategy to deal with small sample size is to use a shrinkage estimator of the covariance matrix, which can be expressed mathematically as

Linear discriminant analysis -- Part 7

Biomedical studies The main application of discriminant analysis in medicine is the assessment of severity state of a patient and prognosis of disease outcome. For example, during retrospective

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analysis, patients are divided into groups according to severity of disease – mild, moderate, and severe form. Then results of clinical and laboratory analyses are studied to reveal statistically different variables in these groups. Using these variables, discriminant functions are built to classify disease severity in future patients. Additionally, Linear Discriminant Analysis (LDA) can help select more discriminative samples for data augmentation, improving classification performance. In biology, similar principles are used in order to classify and define groups of different biological objects, for example, to define phage types of *Salmonella enteritidis* based on Fourier transform infrared spectra, to detect animal source of *Escherichia coli* studying its virulence factors etc.

Linear discriminant analysis -- Part 8

Canonical discriminant analysis for k classes Canonical discriminant analysis (CDA) finds axes ($k - 1$ canonical coordinates, k being the number of classes) that best separate the categories. These linear functions are uncorrelated and define, in effect, an optimal $k - 1$ space through the n -dimensional cloud of data that best separates (the projections in that space of) the k groups. See “Multiclass LDA” for details below. Because LDA uses canonical variates, it was initially often referred as the "method of canonical variates" or canonical variates analysis (CVA).